

CMSE 801 - Day 16

Seaborn

Oct 31, 2022

General Course Announcements

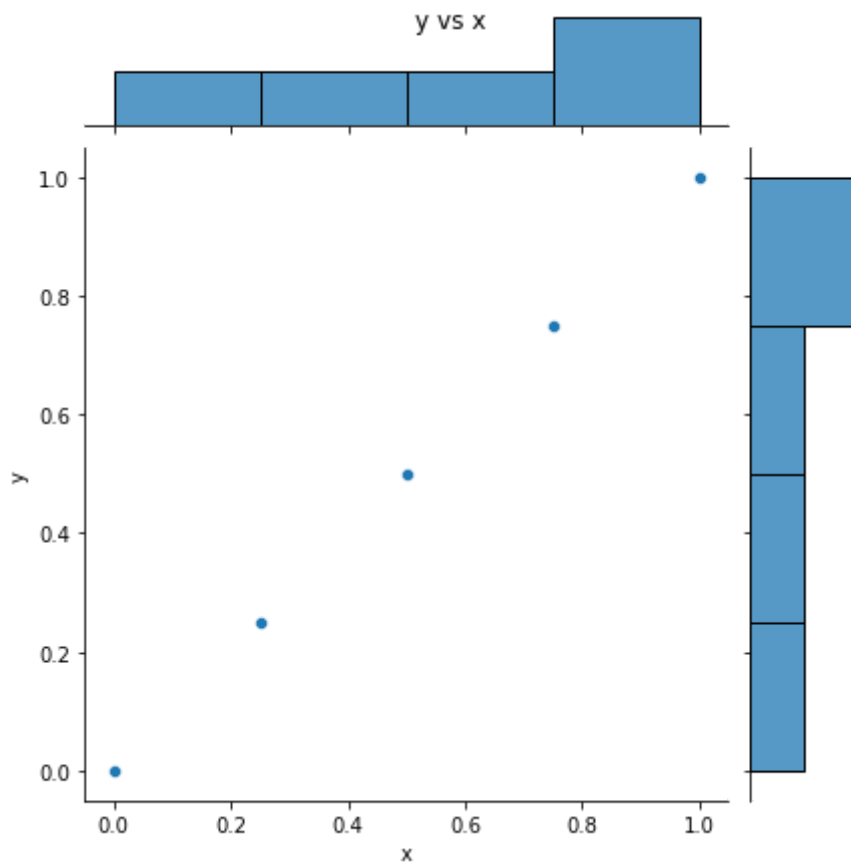
- Homework #3 is due Tuesday Nov 1 at 11:59 p.m.
- Homework #4 was released and is due Monday Nov 14 at 11:59 p.m.
- Homework #5 will be optional for extra credit

Questions/comments from Day 16 PCA

- Could you show how you combined 'ont, eri, sup, mhu' files and plot jointplot?
- I couldn't figure out how to set labels, titles and figure size for sns.jointplot.
- In part 1, when I tried seaborn.jointplot using the argument 'kind="reg"', I met the error saying 'regplot() got an unexpected keyword argument 'hue''. Why?
- It seems that the code for pyplot can also be used in seaborn, but I do not find a way to set xlabel and ylabel jointplot function.
- maybe going over how to add labels (very different than with matplotlib)

```
In [1]: import seaborn as sns
v = [0.0, 0.25, 0.5, 0.75, 1]
w = [0.0, 0.25, 0.5, 0.75, 1]
fig = sns.jointplot(x=v, y=w)
fig.ax_joint.set_xlabel('x')
fig.ax_joint.set_ylabel('y')
fig.fig.suptitle('y vs x')
```

```
Out[1]: Text(0.5, 0.98, 'y vs x')
```



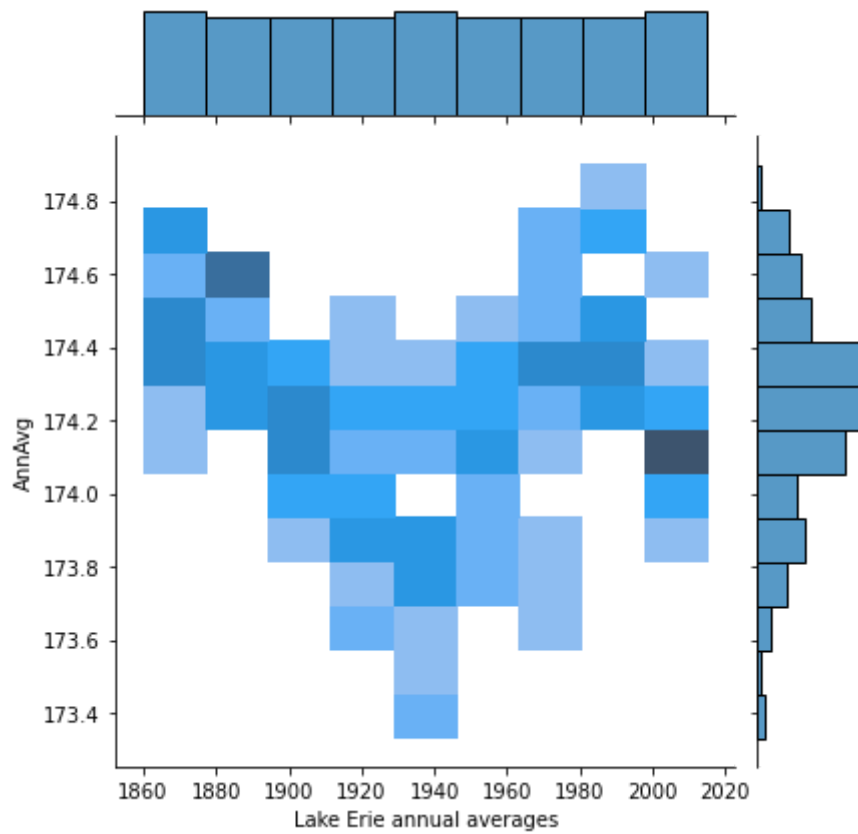
```
In [2]: import pandas as pd
eri = pd.read_csv("eri.csv")
eri = eri.drop(columns=["Unnamed: 2", "Unnamed: 3", "Unnamed: 4"])
eri = eri.rename(columns={"Unnamed: 1": "AnnAvg"})
eri.head()
```

```
Out[2]:
```

	Lake Erie annual averages	AnnAvg
0	1860	174.688483
1	1861	174.712317
2	1862	174.744983
3	1863	174.655733
4	1864	174.471567

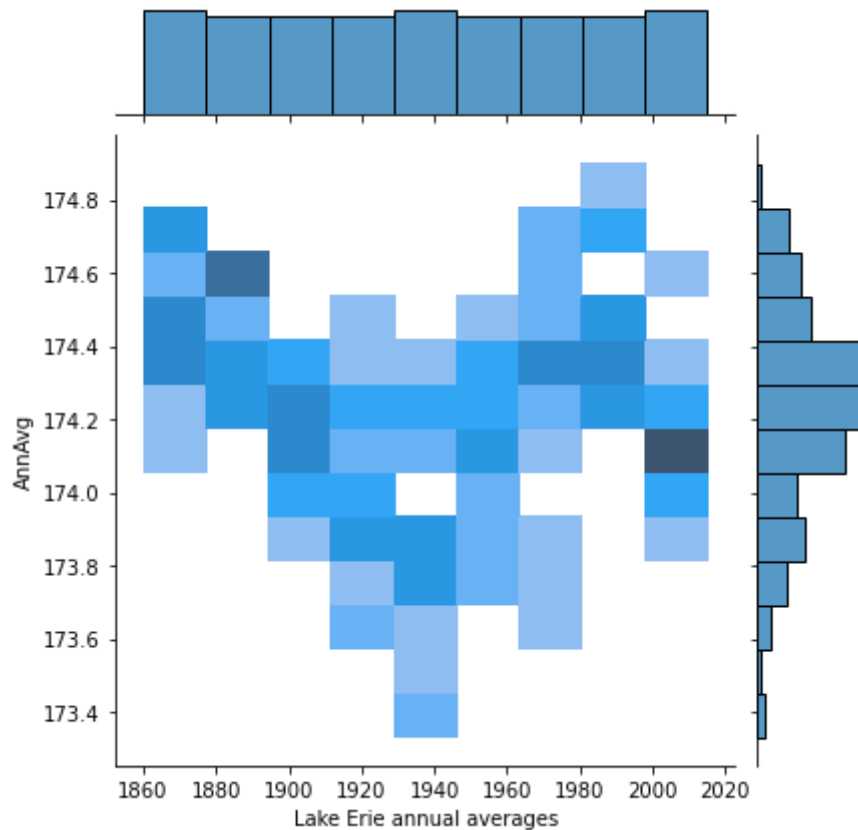
```
In [3]: sns.jointplot(x=eri['Lake Erie annual averages'], y=eri['AnnAvg'], kind='hist')
```

```
Out[3]: <seaborn.axisgrid.JointGrid at 0x7fa5f2c611f0>
```



```
In [4]: sns.jointplot(x=eri.iloc[:,0], y=eri.iloc[:,1], kind='hist')
```

```
Out[4]: <seaborn.axisgrid.JointGrid at 0x7fa5f6605ac0>
```



```
In [5]: sns.jointplot(x=eri.iloc[:,0], y=eri.iloc[:,1], kind='reg')
```

Out[5]: <seaborn.axisgrid.JointGrid at 0x7fa5f6db8850>

